GEMMA-WM: A Hierarchical Sequential JEPA World Model for Global Human-Mobility Modelling

Rationale, survey of world models, architecture, objectives, and pre-registered experimental plan

Technical design note for the GEMMA project
(Global Embeddings for urban Mobility Modelling and Adaptation)
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Status and honesty note. Every number in Section 2 was measured in our own runs or taken from the published Massive-STEPS benchmark [1]. *No result for the new model in this note has been measured yet.* Section 8 states pre-registered *targets* and the criteria under which each hypothesis is rejected. A paper must report whatever the runs produce, including negative ablations.

1 Executive summary

- **Goal.** Pretrain GEMMA once on large source cities so that next-POI recommendation in *unseen* cities (zero-, few- and full-shot) beats in-city supervised models (STHGCN [24], GETNext [25], LSTPM [26], Flashback [27]) and zero-shot LLM agents (LLM-Mob [28], LLM-ZS [29], LLM-Move [30]).
- **What we learned.** The original JEPA pretraining did not improve the backbone once the model was trained to predict the next check-in on the source cities: a randomly initialised backbone reached the same Tokyo accuracy (zero-shot 0.236 vs. 0.234). Gains so far come from user history and the realization stage, and almost none from predicting places a user has never visited ("explore" Acc@1 ≤ 0.01).
- **Diagnosis.** The old JEPA had (i) a single latent target dominated by the POI-name embedding, (ii) a one-step horizon, (iii) a predictor that never had to reason about *decisions*, and (iv) its latent was not the space in which retrieval happens, so pretrained knowledge had no direct path to the ranking.
- **Proposal: HS-JEPA (Hierarchical Sequential JEPA).** One self-supervised world model that predicts, in latent space and coarse-to-fine, the *activity* $\rightarrow$ *primary category* $\rightarrow$ *secondary category* $\rightarrow$ *place* $\rightarrow$ *return-or-explore* of the next H check-ins, against *hierarchical abstraction targets* produced by an EMA target encoder. Retrieval is performed *in the world model's latent space* by an exact hierarchical return–explore mixture (Section 7). Pretraining and retrieval become a single stage.
- **Code.** `gemma/hsjepa.py` (model and losses), `gemma/hrem.py` (latent retrieval), `scripts/run_wm.sh` (pretraining, evaluation, ablations).

2 Where we stand: measured results

Source-city validation. After pretraining on the six source cities, validation Acc@1 is 0.155 (v4 JEPA init), 0.165 (v5 JEPA init) and 0.162 (random init). At initialisation the JEPA backbones start ahead (0.107/0.110 vs. 0.087, next-activity 0.43 vs. 0.07), so JEPA learned useful structure, but the next-check-in objective on the same data recovers it.

Zero-shot vs. LLMs. Under the LLM protocol (200 users, 100 nearest POIs with the truth inserted), the earlier 15-city run beat the best LLM-Move result in 6 of 9 held-out cities; Tokyo zero-shot 0.271–0.282 vs. 0.255. Full-shot averaged 81% of STHGCN and was above it only in Tokyo, within noise.

Benchmark facts that shape the design.

- The current-trajectory prefix averages only 1.5–2.0 check-ins, so the user's history carries most of the signal.

Table 1: Tokyo (never used in pretraining), Massive-STEPS test split, ranking over all 2,982 city POIs. "Revisit" = target already in the user's history or prefix (52–53% of cases); "explore" = new to the user. GEMMA runs use the v5c realizer; zero-shot uses no Tokyo training data.

Model	Shot	Acc@1	NDCG@5	Revisit Acc@1	Explore Acc@1	Next-act. acc
GEMMA v4 (old pipeline)	full	0.107	0.170	–	–	0.364
GEMMA v5c, JEPA backbone v4	zero	0.234	0.330	0.451	0.002	0.502
GEMMA v5c, JEPA backbone v4	full	0.242	0.345	0.445	0.010	0.529
GEMMA v5c, JEPA backbone v5	zero	0.218	0.313	0.421	0.000	0.487
GEMMA v5c, **random backbone**	zero	0.236	0.326	0.454	0.002	0.508
GEMMA v5c, **random backbone**	full	0.243	0.344	0.447	0.010	0.509
History-frequency heuristic (0 parameters)	–	0.175	0.296	–	0.000	–
Activity heuristic (frequency + Markov)	–	–	–	–	–	0.530
STHGCN [24] (published)	in-city	0.250	0.350	–	–	–
LSTPM [26] (published)	in-city	0.225	0.315	–	–	–

- 40–62% of targets are already in the user's history; the rest are explorations.
- The best LLM numbers rank roughly 101 candidates with the truth inserted, not the full vocabulary.

3 Why a world model for global mobility

Mobility as a partially observed decision process. A user's day is a sequence of decisions (*what to do*, *where*, *return or try something new*) generated by a latent state: routine, purpose, spatial anchors, time budget. A world model learns $p(s_{t+1} \mid s_{\leq t})$ over such latent states. Prediction *in latent space* (JEPA [3, 4, 5]) rather than over POI tokens has four properties that matter for *global* pretraining:

1. **Transfer.** Venue identities are city-specific (tens of thousands per city) and do not transfer. Latent dynamics do: activity grammar, daily motifs [32], distance decay, periodicity, exploration–return balance [31].
2. **Ignoring the unpredictable.** Which of several equivalent cafés a user picks is close to irreducible noise. A latent target can encode "a café near the office at 9 am" without being penalised for the exact venue, whereas a token-level generative objective spends capacity on that noise [3].
3. **Hierarchy matches behaviour.** Travel-behaviour theory models choice as nested decisions (activity, then destination; nested logit and activity-based models [33, 34]). A hierarchical latent predictor is the neural, self-supervised counterpart of that nesting.
4. **Planning and counterfactuals.** A latent world model can roll forward several steps (next H check-ins) and score hypothetical futures, which pure retrieval models cannot.

Why our first JEPA did not help, and what changes.

Failure in GEMMA v4/v5	HS-JEPA remedy
One latent target per check-in, dominated by the LLM name embedding	Four hierarchical abstraction targets (activity, primary, secondary, place) from the same EMA target encoder
One-step horizon	Multi-horizon prediction of the next H=3 check-ins (block-causal)
No decision variable in the latent	Explicit return-or-explore query, read out as an activity-conditioned gate
Pooled-context predictor	Cross-attention predictor over all context states
Retrieval in a separately learned space	Retrieval reads the predicted latents directly (pointer into latent memory, latent–POI matching)
Two stages (JEPA, then next-place)	One stage: JEPA losses and the retrieval likelihood trained jointly from step 0

4 Survey: world models and their fit for mobility

Choice. We keep the JEPA family as the world-model core. We add (a) hierarchical, decision-aligned latent targets, (b) multi-horizon sequential prediction, and (c) latent-space retrieval. The semantic-ID autoregressive head stays as an

Table 2: World-model families. "Fit" is our judgement for city-agnostic check-in modelling with huge, city-specific item vocabularies and no actions.

Family	Examples	Predicts	Fit for global mobility
Latent RL world models (RSSM)	World Models [10], PlaNet [11], DreamerV3 [12]	Stochastic latent + reconstruction of observations	Good latent dynamics, but the reconstruction loss forces modelling venue identity; no actions here. Option: stochastic latent as an ablation.
Value-equivalent models	MuZero [13]	Reward/value/policy-relevant latents	Needs rewards and actions; not applicable without a planning task.
Latent MPC models	TD-MPC2 [14]	Latent dynamics + value, no reconstruction	Elegant latent-only training, but action-centric.
Token autoregressive world models	IRIS [15], Genie [16], GAIA-1 [17], Cosmos [18]	Discrete tokens of future observations	Applicable via semantic IDs [23]; spends capacity on identity noise; we keep it only as an auxiliary decoder.
Diffusion world models	DIAMOND [19]	Denoised future observations	Strong for pixels; costly and identity-centric for POIs.
Joint-embedding predictive (JEPA)	I-JEPA [4], V-JEPA [5], V-JEPA 2 [6], MC-JEPA [7], LeJEPA [8]; H-JEPA (concept) [3]	Latent of masked/future content	**Best fit**: latent targets transfer across cities; V-JEPA 2 shows latent predictors can be action-conditioned and used for planning.
Frozen-feature latent dynamics	DINO-WM [20]	Dynamics on pretrained features	Analogue: dynamics over frozen LLM POI embeddings; weaker than learning the encoder.
Trajectory JEPAs	T-JEPA [21], HiT-JEPA [22]	Latents of masked trajectory parts	Closest prior work: built for *trajectory similarity*. HiT-JEPA's hierarchy is spatial granularity (point / intermediate / whole), not semantic decisions.
LLM mobility agents	LLM-Mob [28], LLM-ZS [29], LLM-Move [30]	Next place via prompting	Zero-shot, but billion-parameter and evaluated on small candidate lists.

auxiliary generative decoder, and a stochastic-latent RSSM variant is listed as future work.

Novelty against prior work (to be checked in the related-work pass). H-JEPA is a conceptual proposal in [3]. HiT-JEPA [22] is hierarchical over spatial scale for trajectory similarity. To our knowledge, HS-JEPA is the first JEPA whose hierarchy follows the *semantic decision hierarchy* (activity → category → place → return/explore) with abstraction-level targets, and the first to perform next-POI retrieval inside the JEPA latent space across cities.

5 HS-JEPA: architecture

Tokens. Each check-in token combines the LLM embedding of "category : name", the 4-level semantic ID (RQ-VAE), activity/primary/secondary ids from the Foursquare taxonomy, multi-scale Fourier time (hour, weekday, month), and relative motion (log-distance, bearing). All features are city-agnostic.

Context encoder. The online encoder f_θ (4-layer Transformer, d=256) reads the diary $c = [\text{trail}_1]\,\texttt{SEP}\cdots\texttt{SEP}\,[\text{prefix}]$ and outputs states $h \in \mathbb{R}^{L\times d}$.

Hierarchical sequential predictor. For horizons $h = 1..H$ and levels $k \in \{\text{act}, \text{pri}, \text{sec}, \text{place}, \text{ret}\}$, a query $q_{h,k} = [\text{MASK}] + \text{pos}(n{+}h) + e^{\text{lvl}}_k + e^{\text{hor}}_h + \bar{h}$ cross-attends to h. Under a block-causal mask, $q_{h,k}$ may attend to $q_{h',k'}$ if $h' < h$, or if $h' = h$ and $k' \le k$:

$$z_{h,k} = g_\phi\left(q_{h,k} \mid h;\ z_{<h,\cdot},\, z_{h,<k}\right). \tag{1}$$

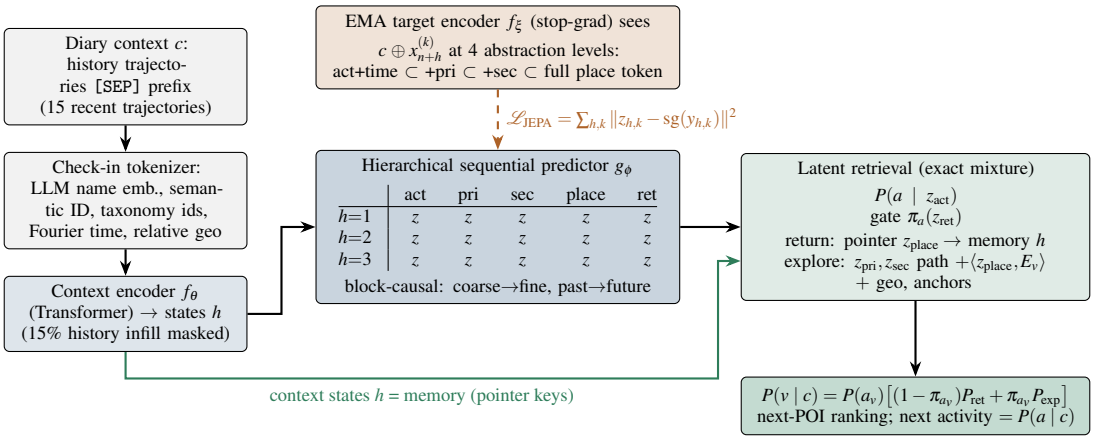


Figure 1: GEMMA-WM. One self-supervised model: context encoder, hierarchical sequential predictor trained against hierarchical abstraction targets (JEPA), and retrieval performed directly on the predicted latents.

The place latent is therefore predicted *conditioned on the predicted activity and category latents*, the sequential hierarchy, and step $h{+}1$ is conditioned on the predicted step h, a latent rollout.

Hierarchical abstraction targets. The EMA target encoder f_ξ ($\xi \leftarrow \tau\xi + (1-\tau)\theta$) sees the full context plus the true future check-in at four levels of detail: $x^{(\text{act})}$ = activity id + time; $x^{(\text{pri})}$ = + primary id; $x^{(\text{sec})}$ = + secondary id; $x^{(\text{place})}$ = the full token:

$$y_{h,k} = \text{LN}\big(f_\xi(c \oplus x^{(k)}_{n+h})\big)_{n+h}, \qquad \mathscr{L}_{\text{JEPA}} = \frac{1}{4}\sum_{k \le \text{place}} \frac{1}{|\mathscr{V}|}\sum_{h \in \mathscr{V}} \|z_{h,k} - \text{sg}(y_{h,k})\|_2^2. \tag{2}$$

Each level must encode exactly the information of its abstraction, so the latent space becomes hierarchical by construction rather than by hope.

6 Self-supervised objectives and masking

All targets come from the check-in stream itself; no human labels are used.

$$\mathscr{L} = \underbrace{-\log P(v^\star \mid c)}_{\text{latent retrieval}} + 0.3\,\underbrace{\text{BCE}(\pi_{a^\star}, \mathbb{1}[v^\star \text{ new}])}_{\text{return/explore}} + \lambda_J \mathscr{L}_{\text{JEPA}} + \lambda_I \mathscr{L}_{\text{infill}} + \lambda_P \sum_{h,\ell \in \{a,p,s\}} \text{CE}(\ell_h) + \lambda_R\,\text{VICReg}(h) \tag{3}$$

Defaults: $\lambda_J{=}1$, $\lambda_I{=}0.5$, $\lambda_P{=}0.5$, $\lambda_R{=}0.1$, $H{=}3$; SIGReg [8] is an option in place of VICReg [9]. A light semantic-ID decoder loss keeps the generative head [23] trained on $z_{1,\text{place}}$.

Masking strategy. The masking defines what the world model must predict, so it should rehearse the downstream question without making it trivial:

Strategy	What it teaches	Used?
Future block (next H)	Dynamics and multi-step rollout; exactly the downstream query	**Yes** (main view)
Hierarchical abstraction	Coarse-to-fine decision structure; transferable activity/category grammar	**Yes** (targets)
History infill (15%)	Representation of the user's routine and memory (bidirectional)	**Yes** (second view)
Random span / activity block (v4)	Generic interpolation; leaks future context into the input	Ablation only
Whole-trajectory masking	Day-level routine (motif) prediction	Future work

7 World-model retrieval

Retrieval reads the predicted latents of step h=1:

$$P(v \mid c) = P(a_v \mid c)\Big[(1-\pi_{a_v}(c))\,P_{\text{ret}}(v \mid c, a_v) + \pi_{a_v}(c)\,P_{\text{exp}}(v \mid c, a_v)\Big]. \tag{4}$$

- **Activity** $P(a \mid c)$: probe on $z_{1,\text{act}}$ plus the user's activity profile and activity transitions. It therefore contains the heuristic signals, and the next-activity prediction is this distribution.
- **Gate** $\pi_a(c) = \sigma\big(w^\top z_{1,\text{ret}} + \beta_1 \operatorname{logit} \nu_{u,a} + \beta_2 \log S_u + \dots\big)$. Here $\nu_{u,a}$ is the user's first-visit rate in activity a, and $\beta_2 \log S_u$ is the EPR law [31] $P(\text{explore}) \propto S_u^{-\gamma}$ in logit form, so the learned β_2 is an estimate of $-\gamma$.
- **Return** P_{ret}: softmax over visited POIs of activity a. It combines a latent pointer $\langle W_q z_{1,\text{place}}, W_k h_j \rangle$ (the world model reading its memory) with recency, time-of-day and weekday affinity (as in Flashback [27]), personal transitions and distance.
- **Explore** P_{exp}: softmax over unvisited POIs of activity a. It combines the predicted category path $\log p(p_v \mid z_{\text{pri}}) + \log p(s_v \mid z_{\text{sec}})$, the latent match $\tau \cos(z_{1,\text{place}}, E_v)$ with the POI's semantic-ID/LLM embedding E_v, distance from the last check-in, and distance to the user's anchors. Target-city transition, popularity and time-of-day statistics are added only in few/full-shot.

The mixture is exactly normalised (verified $\sum_v P(v \mid c) = 1$). Nothing is pruned, so there is no recall ceiling, and new places can win when the gate says the user is exploring. This explicitly targets the explore Acc@1 ≈ 0 failure in Table 1.

8 Experimental plan and pre-registered hypotheses

Protocol.

- Massive-STEPS splits; ranking over all city POIs.
- The LLM protocol is reported separately: 200 users, 5 seeds, three candidate sets.
- 9 held-out cities for true zero-shot (or a 3-fold city split so all 15 are held out); 3 seeds.
- Baselines: STHGCN, GETNext, LSTPM, Flashback (published, and re-run on three cities); LLM-Mob, LLM-ZS, LLM-Move; history and activity heuristics; and the same architecture trained only on the target city.

Table 3: Hypotheses, measurements and rejection criteria. "Target" values are goals, not results.

Hypothesis	Measured by	Target / rejected if
H1: HS-JEPA pretraining helps transfer	zero-shot and 5% few-shot vs. the same model with λ_J=λ_I=0	Target $\geq$+0.02 Acc@1 on held-out cities; rejected if the gap is within one SE
H2: the hierarchy matters	full model vs. `--wm_flat`	Target: higher explore Acc@5 and next-activity; rejected if equal
H3: multi-horizon helps	H=3 vs. H=1	Rejected if equal within one SE
H4: global pretraining beats in-city learning	GEMMA vs. the same model trained only on the target	Target: largest gain in small cities; rejected if no gain
H5: the mixture unlocks exploration	explore Acc@1/@5 vs. v5c (0.010 / 0.051 in Tokyo)	Target explore Acc@1 $\geq$ 0.03
H6: next activity	vs. frequency+Markov heuristic (0.530 in Tokyo)	Target $\geq$ heuristic in every city
H7: overall	Acc@1/NDCG@5 vs. STHGCN and LLM-Move (matched protocols)	Target: above STHGCN in a majority of cities and above LLM-Move in all held-out cities, with 3-seed CIs
H8: the gate recovers mobility law	learned β_2 across cities	Consistent negative β_2 ($\gamma > 0$)

Scaling. The strongest evidence for a foundation model is a curve: held-out zero-shot accuracy versus pretraining data, from 1 to 6 source cities and then to Semantic-Trails scale (STD 2013/2018 cover thousands of cities [2]; the 15 benchmark cities must be excluded).

9 Risks and honest limitations

- **The JEPA terms may still add nothing when next-place data is abundant.** H1 tests exactly this; if rejected, the paper reports it and positions HS-JEPA's value at small data or out-of-distribution cities, where the measured initialisation advantage (Section 2) suggests it may matter.
- **Compute.** Four target-encoder passes per step, about $2\times$ the cost of the v5 realizer; pretraining takes several hours per run on one 24 GB GPU.
- **Protocol dependence.** LLM baselines are taken from published tables. Their candidate construction is inferred, so we report all three variants and plan to re-run LLM-Move.
- **Non-causal history** in the random benchmark split is shared with all baselines; we report a causal-history ablation.

10 Implementation map

File	Content
`gemma/hsjepa.py`	`HPredictor` (block-causal hierarchical predictor), `WMHead`, `wm_forward` (online + 4 target passes, infill, probes, regulariser)
`gemma/hrem.py`	`hrem_mixture`: exact activity $\rightarrow$ gate $\rightarrow$ return/explore retrieval used by the world model
`gemma/realize.py`	City index, user-history evidence, training loop (EMA hook, AMP, mid-epoch validation), evaluation
`scripts/train_realizer_source.py`	Source-city pretraining (`--retriever wm`)
`scripts/run_downstream.py`	Zero/few/full evaluation, both protocols
`scripts/run_wm.sh`	Pretraining from scratch, Tokyo evaluation, old-JEPA-initialised variant, ablations (`ABLATIONS=1`), 15 cities (`ALL_CITIES=1`)

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